

Same probe, different depths: how the gap between output and internals scales in biological foundation models

Ihor Kendiukhov*

Institute of Medical Genetics and Applied Genomics,
University of Tübingen, Tübingen, Germany

Pivotal Research

22 August 2026

Abstract

A language model can hold information internally that its outputs do not express. For biological foundation models — protein and genome language models now used in enzyme design and variant effect prediction — the size of that discrepancy bears on whether output-only evaluation is a sufficient safety practice. We define an **elicitation gap**: the difference between what a model’s sanctioned output expresses and what a linear readout recovers from its best internal layer, both read at *identical capacity*, so that a difference is a property of the model rather than of the probe. We measure how it scales across five tasks, two modalities and four model families, repeating every experiment up to six times.

The gap widens with scale on some tasks and not others: clearly on fold recognition (+0.020 per decade of parameters, 95% CI [+0.009, +0.031]) and enzyme function (+0.021, [+0.014, +0.029]); not detectably on two ceiling-compressed transcription-factor tasks; and in the opposite direction on a genome task (−0.019, [−0.037, −0.001]). Using a measurement that holds the score fixed and varies the budget, we further find that the budget required to reach a fixed capability falls with scale (−0.118 decades of budget per decade of parameters, [−0.181, −0.055]).

Our central methodological finding is that single-run scaling claims in this literature are unreliable in both directions. Our own headline reversed under replication: one run gave “no measurable change” (+0.018, [−0.011, +0.046]); six runs gave a clear widening (+0.020, [+0.009, +0.031]). The point estimate barely moved — only the interval did. Of twenty-nine measurement defects we found in our own pipeline, every one of the twenty-six that biased a number biased it in our favour, an asymmetry we argue is structural rather than incidental.

1 Introduction

As models get larger, does the gap between what they say and what they internally represent widen or close?

If it closes, output-only assessment becomes progressively safer as the field advances: what you see is increasingly what the model has. If it widens, output-only assessment becomes progressively *more* misleading exactly as models become more capable — and for biological models the capability

*Correspondence: ihor.kendiukhov@student.uni-tuebingen.de

in question includes designing enzymes, predicting variant effects, and reasoning about viral fitness.

The question is not whether internal states contain more than outputs. For protein models that is established: structural and functional information emerges inside representations as pretraining scales [37], the best internal layer outperforms the final layer on downstream tasks, with the deepest layer winning in only 17.92% of model–task pairs across 13 protein language models and 15 tasks [23], and medium-sized models transferring as well as large ones on realistic datasets [46]. The question is whether that discrepancy is a fixed property of the architecture or a function of scale, and that requires measuring it the same way across a ladder of model sizes. We report slopes per decade of parameters, following the convention of the scaling-law literature [24], while measuring a different quantity: not loss against scale, but the difference between two ways of reading one model.

A second concern runs through this paper. Seed and split variance in machine-learning benchmarks is large enough that single-run comparisons routinely mistake noise for effect [4], reported orderings can reverse under reseeding alone [16], and fine-tuned scores vary substantially with initialisation and data order [11]. We therefore repeat every measurement and report intervals [30].

1.1 Related work

Capability elicitation. That behavioural evaluation understates what a model can do is established outside biology. Password-locked models show that fine-tuning recovers capabilities that prompting does not [14]; a direct comparison of prompting, activation steering and fine-tuning concludes that fine-tuning should be the method of choice for trustworthy capability evaluation [20]; and internal representations have been shown to encode the correct answer while the output is wrong [35]. Our B/P/F design is the closest analogue of that work in the biological modality, and our contribution is to ask how the discrepancy behaves *as a function of scale*.

Probing methodology. Reading a linear classifier off intermediate layers is an established protocol [2], and its central difficulty is known: a sufficiently expressive probe can learn the task itself, so a probe score says as much about the probe as about the model [3]. Hewitt and Liang [18] make this precise with control tasks, arguing that a probe is interpretable only when its capacity is constrained and benchmarked against a random-label control; Voita and Titov [47] instead replace accuracy with description length. We take the first route and hold capacity fixed across surfaces, which is what makes the gap a difference between two readings of one model rather than a comparison of two probes.

Protein model benchmarks. Benchmarks in this field are broad rather than deep. TAPE [36], PEER [48], FLIP [10], ProteinGym [33], ProteinWorkshop [22] and PFMBench [13] together cover many dozens of tasks. Closest to our P surface, Unsal et al. [45] benchmark twenty protein representation methods on four function-centric tests — semantic similarity, ontology-based function prediction, drug-target family classification and protein–protein binding affinity — establishing that representation quality is the right unit of comparison for these models. That work compares methods at one representation surface each; the capacity-matched B/P contrast here is designed to extend that comparison to two surfaces of the same model, across a scale ladder. PFMBench in particular runs the full ESM2 ladder on 11 tasks. What they generally do not report is a repeat: a single run per model size, with no interval or a best-minus-worst spread over

about three runs, is the norm. Section 3.5 is about what that costs. FLIP additionally documents a task on which a one-hot linear regression ties a pretrained transformer, and CARE [49] shows a 2024 method beating BLASTp by only 55.1% to 51.4% on its hardest split — both reasons to require a stated baseline. Comparing frozen-feature linear evaluation against fine-tuning across a scale-ordered set of models has a template outside biology [26], which bears on our P-versus-F contrast.

Scale in protein language models. Whether bigger is better here is already contested. Ankh reports that ESM2-15B did not outperform smaller ESM models on all tasks and showed inconsistent results across runs [12]; reverse-distillation work reports the ESM2 family peaking at 650M–3B with 15B degraded [6]; Cheng et al. [7] study compute-optimal training directly; Li et al. [27] analyse transfer as a function of scale; and Serrano et al. [40] ask whether protein language models are compute optimal. Outside biology, compute-optimal training fixes one of budget or performance and reads off the other [19] — the construction we borrow when inverting the cost measurement — while apparent scaling behaviour has been shown to depend on the metric chosen [38]. Our measurement is of a different quantity — not downstream score, but the *difference* between two ways of reading the same model.

Genomic models. Genomic language models now span a wide parameter range [31], and the benchmark suites used to evaluate them themselves read frozen embeddings through a fixed-capacity linear head [15, 29], so on the genome side the probe surface is standard practice rather than an unusual instrument. Their representational power has recently been questioned directly [43], which is context for the reversal we report.

Single-cell models. The critical literature here is the closest in spirit to ours: zero-shot single-cell embeddings underperform simple baselines on clustering and batch integration [25], and deep-learning perturbation prediction does not yet beat linear baselines [1]. We do not contribute a scaling result in this modality, for the structural reason given in Section 4.3.

1.2 Contributions

1. A measurement protocol that reads a model’s sanctioned output and a linear probe of its best internal layer at *identical readout capacity*, so that the difference between them is a property of the model rather than of the instrument, with probe layers selected on validation data and splits held out by homology.
2. An inverted, iso-performance measurement that holds the score fixed and measures the fine-tuning budget required, removing the confound that a fixed budget is worth more to a larger model.
3. An application of both across five tasks, two modalities and four model families, with up to six repeats of every point, showing that the scaling of the gap is task- and architecture-dependent rather than lawful.
4. Evidence that single-run scaling claims in this literature are unreliable in both directions, demonstrated on a headline of our own that reversed under replication.
5. A classification of twenty-nine measurement defects found in our own pipeline, of which every one of the twenty-six that biased a number biased it in our favour.

2 Materials and methods

2.1 Three reading surfaces

Three surfaces are defined (Table 1). B and P are read from every model on every task; F requires an adapter that supports fine-tuning, and is present on 65 of our 235 cards — ESM2 on fold recognition, enzyme function and viral fitness.

surface	what it is
B , behavioural	the model’s sanctioned output — its final-layer representation, used the way the model is meant to be used
P , probe	a linear readout of the best <i>internal</i> layer, chosen on held-out validation data
F , elicited	the score after light parameter-efficient fine-tuning [21] under a fixed, pinned budget

Table 1: The three reading surfaces. B and P are read through the same linear readout at the same width.

The **elicitation gap** is $P - B$; **recoverability** is $F - B$. That fine-tuning recovers performance a frozen readout does not is established for protein models [39]; what we measure is how much it recovers as a function of scale.

The identical-capacity constraint is what makes the gap a fact about the model. B and P are read through the same linear readout at the same width. A gap measured with a larger readout on P than on B would partly measure the probe [3, 18]. Every card records the width each surface was read at, and a regression test asserts they match. An unconstrained-capacity alternative would report description length instead [47]; we hold capacity fixed so that the two surfaces stay directly comparable.

The probe layer is chosen on validation data, never on test. This was not true in an early version and cost us 92% of a headline gap (Section 4.2).

2.2 Result cards and enforced rules

Three of six rules are enforced structurally, in the schema that defines a result card: a card cannot be constructed without at least one baseline, without a permutation null [34], or without either a ceiling or a stated reason none applies. The remaining three — beating the baseline rather than merely carrying one, reporting rank-reversal risk, and pinning data and code — are conventions the tasks follow and the schema does not compel. We state the distinction because “enforced by the schema” and “we always do it” are different guarantees, and only the first survives a careless contributor.

2.3 Homology control

Protein splits are group-disjoint at DIAMOND all-vs-all clusters [5] at 30% identity and 50% coverage; identity clustering of this kind is the standard construction for held-out protein

splits [42]. On fold recognition those clusters are additionally unioned with CATH homologous superfamilies [41], so the held-out set is held out by lineage rather than merely by sequence similarity; the other protein tasks use identity clustering alone.

2.4 Tasks and model ladders

Five tasks that can carry a model ladder, across two modalities, are listed in Table 2. (A third, single-cell, is out of scope for the structural reason given in Section 4.3.) The ladders are ESM2 [28], CARP [50], RITA [17] and Nucleotide Transformer v2 [9].

task	modality	ladder	span	room to improve	seeds
fold recognition (CATH)	protein	ESM2 $\times 5$	2.57 dec	0.51	6
		CARP $\times 4$	3.02 dec		6
		RITA $\times 4$	1.15 dec		4
enzyme function (EC)	protein	ESM2 $\times 5$	2.57 dec	0.63	6
TF identity (human TF)	protein	ESM2 $\times 5$	2.57 dec	0.11	6
TF family (human TF)	protein	ESM2 $\times 5$	2.57 dec	0.19	5 of 6
enhancer classes (NT)	genome	NT-v2 $\times 4$	0.96 dec	0.51	6

Table 2: Tasks, model ladders, ladder span in decades of parameters, headroom and repeat count. The three fold-recognition ladders share one task, so “room to improve” is given once.

“Room to improve” is 1 minus the mean behavioural score **over the rungs and seeds entering that row’s fit**: a gap cannot widen where there is nowhere left to go. Note that both transcription-factor tasks are ceiling-compressed (0.11 and 0.19), not just the first — which bears on Section 3.1.

Ladders exclude the randomly-initialised control arms. Those exist to ask whether a curve reflects learned biology or parameter count; they share a parameter count with their trained twin, so admitting them puts two populations at one x . They were admitted, on both TF tasks, until this paper was checked (Section 4.2).

2.5 Replication and statistical analysis

Experiments are repeated six times where the checkpoints and splits allow. RITA has four seeds, having been added late, and TF family yields five usable of six (Section 4.3). Seeds vary **the train/test split**; the linear readout is a deterministic convex fit and the fine-tuning initialisation is held fixed, so seed variance here is split variance and not initialisation variance. Analyses fit a slope per seed and test the across-seed mean against zero with a Student- t interval. That rule was fixed in writing before the data **for the recoverability test** and applied unchanged to the gap and cost analyses, which were not themselves pre-registered.

2.6 Iso-performance cost measurement

For the cost measurement we invert the dependent variable: we hold the *score* fixed and measure the *budget* required to reach it. Each budget is a separate fine-tuning run annealed to its own endpoint — reading intermediate checkpoints off one long run is biased, by more than the effect under study (Section 4.2). We ran three seeds over five rungs, with nine budgets (25–400 steps)

in the first seed and seven (50–400) in the other two, and evaluated three target scores. A fourth target of 0.55 was run at the first seed and abandoned: three of five models never reached it inside the largest budget, leaving too few uncensored rungs to fit.

2.7 Code and reproduction

Every claim maps to a script and a data directory (Table 3), in the repository named under Data availability. Result cards live under `data/cards/seed0–seed5`, one directory per repeat, and the iso-performance sweeps under `data/sweeps/`. Reproducing every number needs neither a GPU nor model weights: the released cards are the input.

claim	command
Tables 4, 5, Figure 1	<code>python scripts/analyse_gap.py</code>
Table 6	<code>python scripts/analyse_recoverability.py</code>
Table 7, Figure 2	<code>python scripts/analyse_cost.py</code>
CARP width and depth (Section 3.2)	<code>python scripts/analyse_carp_geometry.py</code>
all figures	<code>python scripts/make_figures.py</code>
bibliography	<code>python scripts/build_bib.py</code>
every published number	<code>pytest -q</code>

Table 3: Every claim in this paper, and the command that reproduces it. The test suite re-derives each published value from the released cards and fails if it moves.

3 Results

3.1 Scaling of the elicitation gap across tasks

task	family	seeds	mean slope/decade	95% CI	seeds agreeing
fold recognition	ESM2	6	+0.0202	[+0.0092, +0.0312]	6 of 6
enzyme function	ESM2	6	+0.0211	[+0.0135, +0.0286]	6 of 6
TF identity	ESM2	6	−0.0026	[−0.0086, +0.0034]	disagree
TF family	ESM2	5	−0.0078	[−0.0447, +0.0291]	disagree
enhancer classes	NT-v2	6	−0.0190	[−0.0368, −0.0012]	disagree

Table 4: Elicitation-gap slopes per decade of parameters (`analyse_family_gap.py`).

Table 4 and Figure 1 give the per-seed slopes. Two tasks widen, clearly and reproducibly. Fold recognition and enzyme function share no data, no label space and no split procedure, and all twelve of their seeds are positive. **These are the strongest results we have.** Their means differ by 0.0009, but we do not read that agreement as evidence of a shared constant: the twelve individual seed slopes span +0.006 to +0.040, a factor of seven, so the closeness of the two means is within their own sampling noise. The two also share seed indices and are correlated across paired seeds ($r = 0.59$), so they are two tasks and not two independent experiments.

Two tasks show nothing, and **both are ceiling-compressed** — TF identity has 0.11 of room and TF family 0.19, against 0.51 and 0.63 for the two that widen. A gap cannot open where the

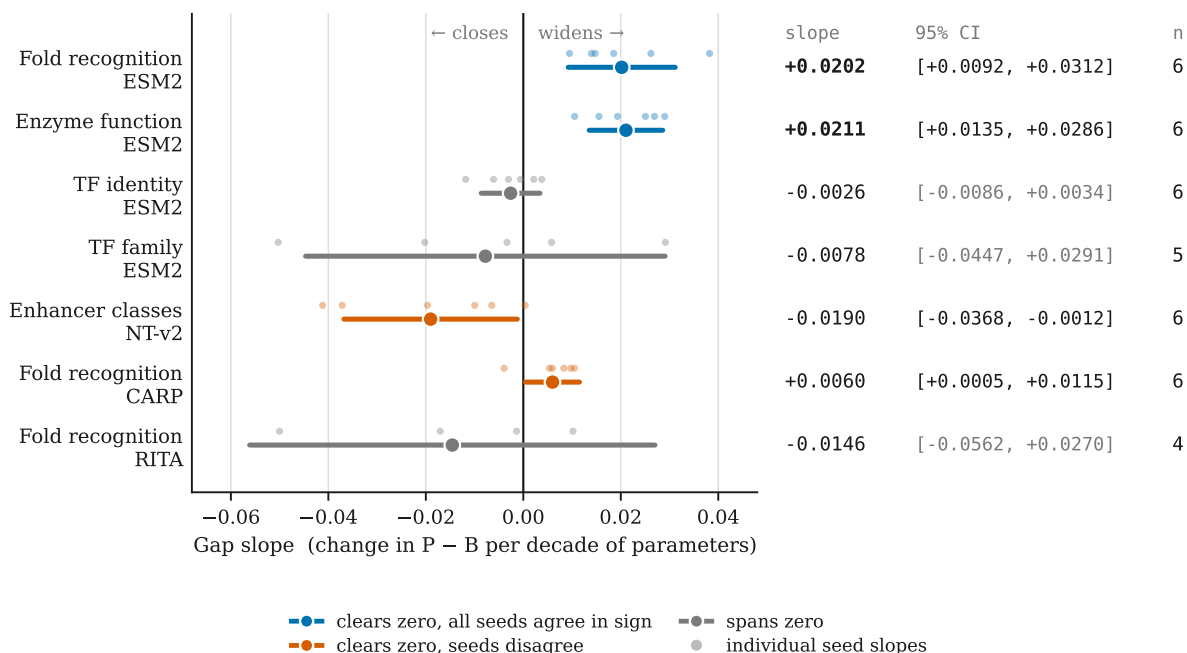


Figure 1: Per-seed slopes of the elicitation gap on $\log_{10}(\text{parameters})$, fitted one slope per seed and averaged across seeds with a Student- t interval. Small dots are individual seed slopes. Only two rows both clear zero and agree in sign across every seed.

behavioural score is already near the maximum, so these two nulls are consistent with ceiling artefacts rather than with an absence of the effect. We cannot distinguish those possibilities with this design.

The genome task points the other way, and its interval clears zero by 0.0012 while its seeds disagree in sign. **We report it as weak.** It has as much room to improve (0.51) as fold recognition (0.51), so headroom does not explain that difference; modality, architecture and family remain confounded, and this design cannot separate them. Its ladder is also the shortest here at 0.96 decades.

The widening is therefore not a general property of biological foundation models. Any claim that it is would overreach the evidence.

3.2 A second and a third model family

Fifty of our sixty-nine protein cards, before CARP was added, were ESM2 — forty-two trained checkpoints and eight randomly-initialised control arms. To test whether the finding is a property of one lab’s recipe we added CARP [50]: four sizes over 3.02 decades, one training set, one objective, and a dilated-convolutional architecture rather than a transformer.

Table 5 sets the three families side by side. CARP agrees in direction and not in magnitude: a third the size, a lower bound clearing zero by 0.0005, one dissenting seed. We report this as **weak corroboration, not confirmation**.

An incidental observation may matter more. CARP’s per-rung gaps are about four times smaller than ESM2’s in absolute terms (mean $|\text{gap}|$ 0.011 against 0.044) and every CARP rung’s six-seed interval straddles zero, while every ESM2 rung’s clears it. And CARP-640M scores higher

family	span	seeds	mean slope/decade	95% CI	seeds agreeing
ESM2	2.57 dec	6	+0.0202	[+0.0092, +0.0312]	6 of 6
CARP	3.02 dec	6	+0.0060	[+0.0005, +0.0115]	5 of 6
RITA	1.15 dec	4	−0.0146	[−0.0562, +0.0270]	disagree

Table 5: Fold recognition across three model families.

than ESM2-3B behaviourally in five of six seeds (mean +0.029, 95% CI [−0.003, +0.061]) with a fifth of the parameters — an interval that spans zero, so we claim comparable performance and not superiority. Two strong families, comparable performance, several-fold different in how much their internals hold back. If this holds, the elicitation gap is at least as much a fact about architecture as about scale. No single-family study could have seen it.

Within CARP we can ask whether width or depth carries this, because its rungs vary the two separately. Paired within-seed contrasts do not resolve it: width alone ($\times 8$ width, equal depth) gives +0.0019 [−0.0219, +0.0257] and depth alone ($\times 2$ depth, equal width) gives +0.0106 [−0.0070, +0.0283], both spanning zero. We report the question as open rather than answered.

A note on availability. We surveyed roughly thirty candidate families to find one second ladder. Almost nothing is built to be a scale ladder: ProGen2 [32] gives its top rung twice the tokens; families change their data mixture at the top; Geneformer [44] crosses a training generation between rungs; most families span under 1.5 decades. **The scarcity of clean scale ladders is itself a finding**, and scaling claims in this literature inherit it whether or not their authors say so.

3.3 Recoverability under light fine-tuning

task	mean slope/decade	95% CI	seeds agreeing
fold recognition	+0.0203	[+0.0076, +0.0330]	6 of 6
enzyme function	+0.0238	[+0.0143, +0.0333]	6 of 6

Table 6: How much light fine-tuning adds over the model’s own output (F − B), six seeds each.

Table 6 gives the slopes: twelve seeds of twelve positive, on two tasks. The two task means differ by 0.0035, which is within their own sampling noise rather than evidence of a shared constant; individual seed slopes span +0.006 to +0.040. The two tasks share seed indices and are correlated, so they are not independent replications.

This result is confounded by construction, and we flagged it as unpublishable for weeks on that basis: every rung receives an identical 400-step budget while the largest model has 375 times the parameters of the smallest, so “a fixed budget is worth more to a bigger model” predicts exactly this. The iso-performance measurement of Section 2.6 was built to remove that confound, and is reported next.

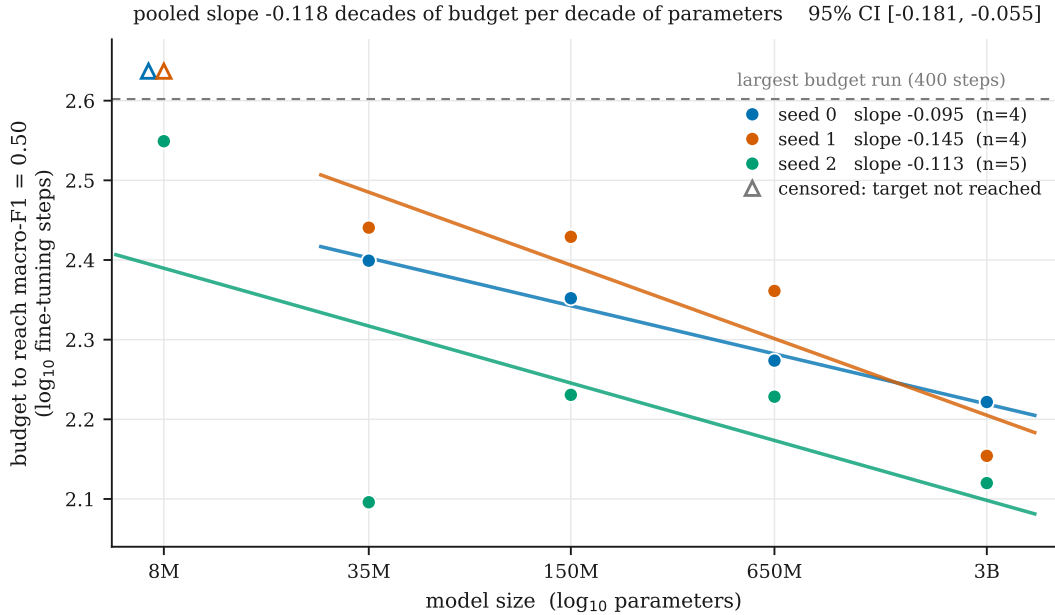


Figure 2: Budget required to reach a fixed macro-F1 of 0.50, against model size. Each budget is a separate fine-tuning run annealed to its own endpoint. Open triangles mark rungs censored at the grid ceiling: that model never reached the target inside the largest budget run, so it is excluded from its seed’s fit.

3.4 Budget required to reach a fixed capability

Applying the iso-performance measurement of Section 2.6 at three target scores gives the slopes in Table 7 and the fits in Figure 2.

target score	mean slope (decades budget / decade params)	95% CI
0.40	-0.040	$[-0.096, +0.016]$
0.45	-0.094	$[-0.265, +0.078]$
0.50	-0.118	$[-0.181, -0.055]$

Table 7: Iso-performance cost scaling, pooled over three seeds.

All nine seed-by-target slopes are negative. At the hardest target the pooled interval clears zero and all three seeds agree in sign: **a tenfold larger model reaches the same score on roughly 0.76 of the budget.**

Three fragilities belong with that number, none of them visible in the table. No individual seed’s own fit clears zero except the first. The pooled interval spans zero if any single seed is dropped — three seeds is the minimum this estimator can speak from, not a comfortable margin. And the smallest rung is censored at this target in two of three seeds but not the third, where it crosses at about 354 of 400 steps; the result depends on that asymmetry, and dropping the smallest rung uniformly across seeds would change it.

This measurement does not carry the fixed-budget confound of the preceding subsection. Two measurements — one confounded, one not — point the same way, which is what makes the

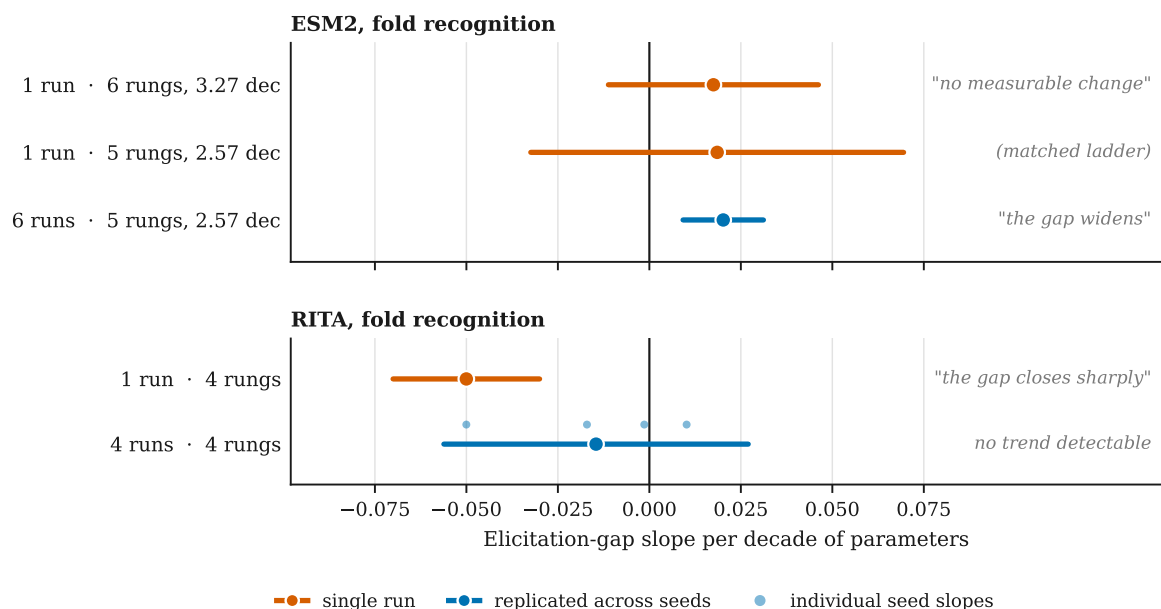


Figure 3: The same quantity, measured once and measured repeatedly. For ESM2 the point estimate barely moves while the interval collapses; for RITA a single run produced an effect clear of zero that four runs erase.

recoverability result interpretable rather than merely suggestive.

The effect appears only where the target is hard enough to separate models. At easier targets every model arrives quickly and the direction, while consistent, is not distinguishable from zero. That is a coherent result rather than a null: capabilities that are cheap for everyone cost roughly the same for everyone.

3.5 Single runs do not survive replication

Our own headline reversed (Table 8, Figure 3):

evidence	ladder	slope	95% CI	what we concluded
1 run of fold recognition	6 rungs, 3.27 dec	+0.018	$[-0.011, +0.046]$	"no measurable change"
1 run, matched ladder	5 rungs, 2.57 dec	+0.019	$[-0.032, +0.069]$	—
6 runs of fold recognition	5 rungs, 2.57 dec	+0.020	$[+0.009, +0.031]$	the gap widens

Table 8: One quantity, measured once and measured six times. The middle row is the like-for-like comparison: the same ladder, read once and read six times.

The middle row is the like-for-like comparison, and it makes the point more sharply than the one we published: on the same ladder, one run gives an interval 4.6 times wider than six runs, around an almost identical estimate.

The measurement barely moved. Only the uncertainty did. The first number was a statement about how little we had run, dressed as a statement about models. We had published it.

It reverses in the other direction too. RITA appeared on a single run to show the gap closing sharply (-0.050 , interval clear of zero, four nearly-collinear points). Four seeds gave -0.050 , -0.001 , -0.017 , $+0.010$: mean -0.015 , interval spanning zero, seeds disagreeing in sign. The apparent effect was one seed.

The same pattern recurred at least six times, on two quantities — the gap and recoverability — across fold recognition, enzyme function, enhancer classification, CARP and RITA. In the ESM2 headline case the estimate barely moved and only the interval did; in the RITA case the estimate itself collapsed to under a third of its single-run value. Both are failures of a single run, by different mechanisms.

The honest form of the claim is narrower than the one we first wrote. **Every single-run result in this programme that we treated as a finding changed its verdict under replication.** The single-run results that survived unchanged were nulls on tasks that are still nulls — which is not reassurance, since a null is what an underpowered design returns by default.

The generalisation is uncomfortable and we state it plainly. Comparable papers in this subfield report either no interval at all or a best-minus-worst spread over about three runs, across four to six model sizes [13, 22, 48]. Three runs is precisely the count that read as a null here before six runs reversed it. The problem is not confined to this subfield: across 485 models and more than a thousand fitted laws, scaling-law estimates are noisy and sensitive to which runs enter the fit [8]. A four- or five-point ladder read once has two or three residual degrees of freedom and an interval wide enough to contain almost anything. “Contains zero” is not “is zero”.

4 Discussion

4.1 Scope of the claims

Several claims are *not* supported by this design, and we set them out explicitly. We do not claim:

- **That the gap widens generally.** It widens on two tasks of five. The distinguishing factor is not identified.
- **That this is about scale rather than architecture.** CARP’s gaps are several-fold smaller than ESM2’s at comparable performance.
- **That four or five checkpoints are a sample of protein language models.** Our intervals reflect seed noise, not model-sampling noise. Strictly, each result is about those checkpoints.
- **Anything about single-cell models.** They have no clean scale ladder.
- **That the cost result holds at easy targets.** It does not.

4.2 Measurement defects and the direction of their bias

The full taxonomy is given in the supplementary material; eight load-bearing ones are given here. We report the classification rather than a tidy count, because the count is the sort of thing this paper is about: of the defects that biased a number at all, **every one biased it in our favour.** The remainder destroyed or corrupted work without pointing in a direction — a failed deploy, an unpinned dependency, a dead parameter. We do not think this asymmetry is luck. **A pipeline is built by people who want it to work, so errors whose output looks like success**

survive review, and errors whose output looks like failure get chased down the same afternoon.

The load-bearing ones:

1. **The probe layer was chosen by argmax on the test set.** On the model where we isolated it (TF identity, ESM2-650M) the honest validation-selected gap is +0.0016 and the test-selected one +0.0188 — a twelvefold inflation, of which $\sim 92\%$ of the apparent gap was selection bias. The fix for it left the same bug on its fallback branch.
2. **An interval that covered 90% while claiming 95%,** which retracted the only trend we had then called significant.
3. **A scaling slope fitted over rungs that failed their own null test** — the p -values were computed by our pipeline, written to the cards, and loaded back by the function doing the fitting, which never read them.
4. **A missing rung was not missing at random.** An out-of-memory error kills the *largest* model, and the top of a ladder is where a plateau masquerading as a trend gets caught. Recovering one such rung turned +0.084 [+0.030, +0.138] into +0.040 spanning zero.
5. **Quantisation reported as an effect.** A budget ratio of exactly 1.00 and a slope of exactly 0.000 were the measuring grid, not the models. *A suspiciously round number is an artefact until shown otherwise.*
6. **A gap of exactly zero because no layer could be chosen** is not a gap of zero. Where a split left too little to carve a validation set, the probe fell back to the sanctioned surface and P equalled B identically — a number that looks like a measurement and is a statement about the instrument.
7. **The measurement built to remove a confound contained one.** Steps-to-target read off a single annealed trajectory is biased, and not even with a constant sign.
8. **Untrained control arms were regressed as if they were models.** Four randomly-initialised ESM2 checkpoints sat inside the ladder on both transcription-factor tasks, at the same parameter counts as their trained twins, silently doubling four of five rungs. This one was found by the adversarial check on *this paper*, after the results in it had been written up. Correcting it did not change any verdict, which is luck rather than vindication.

4.3 Limitations

Model-sampling noise is not represented. Our intervals come from repeated runs on fixed checkpoints. With five checkpoints per family, the population of models is not sampled and cannot be.

Single-cell is out of scope, for a structural reason. Those tasks run one real model plus two non-model comparators. The obvious remedy fails: Geneformer’s three sizes cross a training generation (V1→V2), which is the confound that disqualified other candidates, and V2 alone spans 0.48 decades. No published single-cell ladder holds its recipe fixed.

TF family contributes five usable seeds of six. Its split drops five to eight of fourteen classes depending on seed (seven in four of six), and under one seed too little remained to carve a validation set, so every rung fell back to the sanctioned surface and the whole seed was excluded. The surviving seeds are therefore not a random subset — they are the ones whose inner validation carve retained more than one class. Which way that biases the slope is not established: the excluded seed is middling on class-drop count, so “easier split” is a description we cannot support. More seeds does not fix the selection.

The cost measurement censors its smallest rung at the hardest target, in two of three seeds, leaving four rungs there and five in the third — where that model does reach the target, at about 354 of 400 steps. So it sits at the grid ceiling rather than beyond it, and the “conservative exclusion” argument holds only conditionally: including it steepens the slope if its true cost exceeds roughly 290 steps in the first seed and 380 in the second, and flattens the slope below those thresholds.

Three seeds on the cost measurement is thin. Only its hardest target clears zero, and the pooled interval spans zero if any one seed is removed.

We examined three targets and report one as significant. They are not independent — all three are derived from the same evaluated curves — but the multiplicity is real, and choosing 0.50 as the headline after seeing all three is a forking path we did not pre-register.

5 Conclusion

The empirical results are real, reproducible, and narrower than we would like. Two tasks widen; one measurement free of the obvious confound agrees.

The methodological result is, we think, the more useful contribution. It is cheaply actionable — repeat your runs — it is verifiable against our own published reversals, and it applies to a literature that mostly does not do it. We would rather be the paper that says “our headline reversed, here is the arithmetic” than one more single-run scaling curve.

Key Points

- Reading a model’s sanctioned output and a linear probe of its best internal layer *at identical readout capacity* makes the difference between them a property of the model rather than of the probe.
- The elicitation gap widens with scale on two of five tasks, is undetectable on two ceiling-compressed tasks, and reverses on a genome task; a second architecture shows gaps several-fold smaller at comparable performance, so scaling behaviour here is task- and architecture-dependent rather than lawful.
- Holding the score fixed and measuring the budget required removes the confound that a fixed fine-tuning budget is worth more to a larger model; that budget falls by roughly a quarter per decade of parameters.

- Single-run scaling claims in this literature are unreliable in both directions: our own published headline reversed under replication, with the point estimate almost unchanged and only the interval narrowing.
- Of twenty-nine measurement defects found in our own pipeline, every one of the twenty-six that biased a number biased it in our favour — an asymmetry we argue is structural rather than incidental.

Data availability

All result cards, sweep files and analysis code underlying every number in this paper are openly available at <https://github.com/Biodyn-AI/probe-scaling>. Result cards are under `data/cards/`; iso-performance sweeps under `data/sweeps/`. The analysis scripts, figure-generation script and bibliography generator named in Section 2.7 regenerate every figure, table and reference from those files. No new experimental data were generated; all model checkpoints and benchmark datasets are third-party and cited at first use.

Funding

This work was supported by Pivotal Research.

Declaration of generative AI use

This declaration follows the ISCB acceptable-use policy for large language models.

A large language model (Claude, Anthropic) was used as a coding and checking assistant in the computational work, under the author’s direction and review, in the following ways. It assisted in writing and refactoring the evaluation harness, the analysis functions and the figure-generation code, all of which are released in the repository named under Data availability and are covered by the test suite. It was used as an evaluation technique in the sense the policy describes — to search the pipeline for inconsistencies and anomalies — which is how a number of the measurement defects reported in Section 4.2 were surfaced. It was used to retrieve and verify bibliographic metadata: every reference in this paper was resolved programmatically against doi.org or the arXiv API by the script `build_bib.py`, which fails rather than emit an unresolvable entry. It also assisted in preparing the $\text{\LaTeX}$ sources of the tables and figures.

The scientific claims, their interpretation, and the text of this manuscript are the author’s own. No part of the manuscript was drafted from a prompt. All outputs of the assistant were reviewed by the author, and every numerical result reported here was re-derived from the released result cards by the analysis scripts named in Section 2.7.

References

- [1] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22(8):1657–1661, August 2025. ISSN 1548-7105. doi: 10.1038/s41592-025-02772-6. URL <http://dx.doi.org/10.1038/s41592-025-02772-6>.

- [2] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. URL <https://arxiv.org/abs/1610.01644>.
- [3] Yonatan Belinkov. Probing Classifiers: Promises, Shortcomings, and Advances, 2021. URL <https://arxiv.org/abs/2102.12452>.
- [4] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Naz Sepah, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Dmitriy Serdyuk, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for Variance in Machine Learning Benchmarks, 2021. URL <https://arxiv.org/abs/2103.03098>.
- [5] Benjamin Buchfink, Chao Xie, and Daniel H Huson. Fast and sensitive protein alignment using DIAMOND. *Nature Methods*, 12(1):59–60, November 2014. ISSN 1548-7105. doi: 10.1038/nmeth.3176. URL <http://dx.doi.org/10.1038/nmeth.3176>.
- [6] Darius Catrina, Christian Bepler, Samuel Sledzieski, and Rohit Singh. Reverse Distillation: Consistently Scaling Protein Language Model Representations, 2026. URL <https://arxiv.org/abs/2603.07710>.
- [7] Xingyi Cheng, Bo Chen, Pan Li, Jing Gong, Jie Tang, and Le Song. Training Compute-Optimal Protein Language Models. In *Advances in Neural Information Processing Systems 37*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/8066ae1446b2bbccb5159587cc3b3bcc-Abstract-Conference.html.
- [8] Leshem Choshen, Yang Zhang, and Jacob Andreas. A Hitchhiker’s Guide to Scaling Law Estimation, 2024. URL <https://arxiv.org/abs/2410.11840>.
- [9] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P. de Almeida, Hassan Sirelkhatim, Guillaume Richard, Marcin Skwark, Karim Beguir, Marie Lopez, and Thomas Pierrot. Nucleotide Transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, November 2024. ISSN 1548-7105. doi: 10.1038/s41592-024-02523-z. URL <http://dx.doi.org/10.1038/s41592-024-02523-z>.
- [10] Christian Dallago, Jody Mou, Kadina E. Johnston, Bruce J. Wittmann, Nicholas Bhattacharya, Samuel Goldman, Ali Madani, and Kevin K. Yang. FLIP: Benchmark tasks in fitness landscape inference for proteins. In *Advances in Neural Information Processing Systems 34 Datasets and Benchmarks Track*, 2021. URL <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/2b44928ae11fb9384c4cf38708677c48-Abstract-round2.html>.
- [11] Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping, 2020. URL <https://arxiv.org/abs/2002.06305>.
- [12] Ahmed Elnaggar, Hazem Essam, Wafaa Salah-Eldin, Walid Moustafa, Mohamed Elkerdawy, Charlotte Rochereau, and Burkhard Rost. Ankh: Optimized Protein Language Model Unlocks General-Purpose Modelling, 2023. URL <https://arxiv.org/abs/2301.06568>.

- [13] Zhangyang Gao, Hao Wang, Cheng Tan, Chenrui Xu, Mengdi Liu, Bozhen Hu, Linlin Chao, Xiaoming Zhang, and Stan Z. Li. PFMBench: Protein Foundation Model Benchmark, 2025. URL <https://arxiv.org/abs/2506.14796>.
- [14] Ryan Greenblatt, Fabien Roger, Dmitrii Krashennnikov, and David Krueger. Stress-Testing Capability Elicitation With Password-Locked Models, 2024. URL <https://arxiv.org/abs/2405.19550>.
- [15] Katarína Grešová, Vlastimil Martinek, David Čechák, Petr Šimeček, and Panagiotis Alexiou. Genomic benchmarks: a collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24(1), May 2023. ISSN 2730-6844. doi: 10.1186/s12863-023-01123-8. URL <http://dx.doi.org/10.1186/s12863-023-01123-8>.
- [16] Peter Henderson, Riashat Islam, Philip Bachman, Joelle Pineau, Doina Precup, and David Meger. Deep Reinforcement Learning that Matters, 2017. URL <https://arxiv.org/abs/1709.06560>.
- [17] Daniel Hesslow, Niccoló Zanichelli, Pascal Notin, Iacopo Poli, and Debora Marks. RITA: a Study on Scaling Up Generative Protein Sequence Models, 2022. URL <https://arxiv.org/abs/2205.05789>.
- [18] John Hewitt and Percy Liang. Designing and Interpreting Probes with Control Tasks, 2019. URL <https://arxiv.org/abs/1909.03368>.
- [19] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training Compute-Optimal Large Language Models, 2022. URL <https://arxiv.org/abs/2203.15556>.
- [20] Felix Hofstätter, Teun van der Weij, Jayden Teoh, Rada Djoneva, Henning Bartsch, and Francis Rhys Ward. The Elicitation Game: Evaluating Capability Elicitation Techniques, 2025. URL <https://arxiv.org/abs/2502.02180>.
- [21] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models, 2021. URL <https://arxiv.org/abs/2106.09685>.
- [22] Arian R. Jamasb, Alex Morehead, Chaitanya K. Joshi, Zuobai Zhang, Kieran Didi, Simon V. Mathis, Charles Harris, Jian Tang, Jianlin Cheng, Pietro Lio, and Tom L. Blundell. Evaluating representation learning on the protein structure universe, 2024. URL <https://arxiv.org/abs/2406.13864>.
- [23] Roman Joeres, Ilya Senatorov, Anastasia Kolchina, Dietrich Klakow, and Olga V. Kalinina. Task- and dataset-specific information in protein language models, 2026. URL <https://arxiv.org/abs/2608.12090>.
- [24] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling Laws for Neural Language Models, 2020. URL <https://arxiv.org/abs/2001.08361>.

- [25] Kasia Z. Kedzierska, Lorin Crawford, Ava P. Amini, and Alex X. Lu. Zero-shot evaluation reveals limitations of single-cell foundation models. *Genome Biology*, 26(1), April 2025. ISSN 1474-760X. doi: 10.1186/s13059-025-03574-x. URL <http://dx.doi.org/10.1186/s13059-025-03574-x>.
- [26] Simon Kornblith, Jonathon Shlens, and Quoc V. Le. Do Better ImageNet Models Transfer Better?, 2018. URL <https://arxiv.org/abs/1805.08974>.
- [27] Francesca-Zhoufan Li, Ava P. Amini, Yisong Yue, Kevin K. Yang, and Alex X. Lu. Feature Reuse and Scaling: Understanding Transfer Learning with Protein Language Models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, 2024. URL <https://proceedings.mlr.press/v235/li24a.html>.
- [28] Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, March 2023. ISSN 1095-9203. doi: 10.1126/science.ade2574. URL <http://dx.doi.org/10.1126/science.ade2574>.
- [29] Frederikke Isa Marin, Felix Teufel, Marc Horlacher, Dennis Madsen, Dennis Pultz, Ole Winther, and Wouter Boomsma. BEND: Benchmarking DNA Language Models on biologically meaningful tasks, 2023. URL <https://arxiv.org/abs/2311.12570>.
- [30] Evan Miller. Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations, 2024. URL <https://arxiv.org/abs/2411.00640>.
- [31] Eric Nguyen, Michael Poli, Matthew G. Durrant, Brian Kang, Dhruva Katrekar, David B. Li, Liam J. Bartie, Armin W. Thomas, Samuel H. King, Garyk Brixi, Jeremy Sullivan, Madelena Y. Ng, Ashley Lewis, Aaron Lou, Stefano Ermon, Stephen A. Baccus, Tina Hernandez-Boussard, Christopher Ré, Patrick D. Hsu, and Brian L. Hie. Sequence modeling and design from molecular to genome scale with Evo. *Science*, 386(6723), November 2024. ISSN 1095-9203. doi: 10.1126/science.ado9336. URL <http://dx.doi.org/10.1126/science.ado9336>.
- [32] Erik Nijkamp, Jeffrey A. Ruffolo, Eli N. Weinstein, Nikhil Naik, and Ali Madani. ProGen2: Exploring the boundaries of protein language models. *Cell Systems*, 14(11):968–978.e3, November 2023. ISSN 2405-4712. doi: 10.1016/j.cels.2023.10.002. URL <http://dx.doi.org/10.1016/j.cels.2023.10.002>.
- [33] Pascal Notin, Aaron W. Kollasch, Daniel Ritter, Lood van Niekerk, Steffanie Paul, Han Spinner, Nathan Rollins, Ada Shaw, Rose Orenbuch, Ruben Weitzman, Jonathan Frazer, Mafalda Dias, Dinko Franceschi, Yarin Gal, and Debora S. Marks. ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. In *Advances in Neural Information Processing Systems 36 Datasets and Benchmarks Track*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/cac723e5ff29f65e3fcbb0739ae91bee-Abstract-Datasets_and_Benchmarks.html.
- [34] Markus Ojala and Gemma C. Garriga. Permutation Tests for Studying Classifier Performance. In *2009 Ninth IEEE International Conference on Data Mining*, page 908–913. IEEE, December 2009. doi: 10.1109/icdm.2009.108. URL <http://dx.doi.org/10.1109/ICDM.2009.108>.

- [35] Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs Know More Than They Show: On the Intrinsic Representation of LLM Hallucinations, 2024. URL <https://arxiv.org/abs/2410.02707>.
- [36] Roshan Rao, Nicholas Bhattacharya, Neil Thomas, Yan Duan, Xi Chen, John Canny, Pieter Abbeel, and Yun S. Song. Evaluating Protein Transfer Learning with TAPE, 2019. URL <https://arxiv.org/abs/1906.08230>.
- [37] Alexander Rives, Joshua Meier, Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi Guo, Myle Ott, C. Lawrence Zitnick, Jerry Ma, and Rob Fergus. Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences*, 118(15), April 2021. ISSN 1091-6490. doi: 10.1073/pnas.2016239118. URL <http://dx.doi.org/10.1073/pnas.2016239118>.
- [38] Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are Emergent Abilities of Large Language Models a Mirage?, 2023. URL <https://arxiv.org/abs/2304.15004>.
- [39] Robert Schmirler, Michael Heinzinger, and Burkhard Rost. Fine-tuning protein language models boosts predictions across diverse tasks. *Nature Communications*, 15(1), August 2024. ISSN 2041-1723. doi: 10.1038/s41467-024-51844-2. URL <http://dx.doi.org/10.1038/s41467-024-51844-2>.
- [40] Yaiza Serrano, Álvaro Ciudad, and Alexis Molina. Are Protein Language Models Compute Optimal?, 2024. URL <https://arxiv.org/abs/2406.07249>.
- [41] Ian Sillitoe, Nicola Bordin, Natalie Dawson, Vaishali P Waman, Paul Ashford, Harry M Scholes, Camilla S M Pang, Laurel Woodridge, Clemens Rauer, Neeladri Sen, Mahnaz Abbasian, Sean Le Cornu, Su Datt Lam, Karel Berka, Ivana Hutařová Varekova, Radka Svobodova, Jon Lees, and Christine A Orengo. CATH: increased structural coverage of functional space. *Nucleic Acids Research*, 49(D1):D266–D273, November 2020. ISSN 1362-4962. doi: 10.1093/nar/gkaa1079. URL <http://dx.doi.org/10.1093/nar/gkaa1079>.
- [42] Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35(11):1026–1028, October 2017. ISSN 1546-1696. doi: 10.1038/nbt.3988. URL <http://dx.doi.org/10.1038/nbt.3988>.
- [43] Ziqi Tang, Nirali Somia, Yiyang Yu, and Peter K. Koo. Evaluating the representational power of pre-trained DNA language models for regulatory genomics. *Genome Biology*, 26(1), 2025. ISSN 1474-760X. doi: 10.1186/s13059-025-03674-8. URL <http://dx.doi.org/10.1186/s13059-025-03674-8>.
- [44] Christina V. Theodoris, Ling Xiao, Anant Chopra, Mark D. Chaffin, Zeina R. Al Sayed, Matthew C. Hill, Helene Mantineo, Elizabeth M. Brydon, Zexian Zeng, X. Shirley Liu, and Patrick T. Ellinor. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, May 2023. ISSN 1476-4687. doi: 10.1038/s41586-023-06139-9. URL <http://dx.doi.org/10.1038/s41586-023-06139-9>.
- [45] Serbulent Unsal, Heval Atas, Muammer Albayrak, Kemal Turhan, Aybar C. Acar, and Tunca Doğan. Learning functional properties of proteins with language models. *Nature Machine*

- Intelligence*, 4(3):227–245, March 2022. ISSN 2522-5839. doi: 10.1038/s42256-022-00457-9. URL <http://dx.doi.org/10.1038/s42256-022-00457-9>.
- [46] Luiz C. Vieira, Morgan L. Handojo, and Claus O. Wilke. Medium-sized protein language models perform well at transfer learning on realistic datasets. *Scientific Reports*, 15(1), 2025. ISSN 2045-2322. doi: 10.1038/s41598-025-05674-x. URL <http://dx.doi.org/10.1038/s41598-025-05674-x>.
 - [47] Elena Voita and Ivan Titov. Information-Theoretic Probing with Minimum Description Length, 2020. URL <https://arxiv.org/abs/2003.12298>.
 - [48] Minghao Xu, Zuobai Zhang, Jiarui Lu, Zhaocheng Zhu, Yangtian Zhang, Ma Chang, Runcheng Liu, and Jian Tang. PEER: A Comprehensive and Multi-Task Benchmark for Protein Sequence Understanding. In *Advances in Neural Information Processing Systems 35 Datasets and Benchmarks Track*, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/e467582d42d9c13fa9603df16f31de6d-Abstract-Datasets_and_Benchmarks.html.
 - [49] Jason Yang, Ariane Mora, Shengchao Liu, Bruce J. Wittmann, Anima Anandkumar, Frances H. Arnold, and Yisong Yue. CARE: a Benchmark Suite for the Classification and Retrieval of Enzymes, 2024. URL <https://arxiv.org/abs/2406.15669>.
 - [50] Kevin K. Yang, Nicolo Fusi, and Alex X. Lu. Convolutions are competitive with transformers for protein sequence pretraining. *Cell Systems*, 15(3):286–294.e2, March 2024. ISSN 2405-4712. doi: 10.1016/j.cels.2024.01.008. URL <http://dx.doi.org/10.1016/j.cels.2024.01.008>.